HÖRMANN

A MATH ADVENTURE

REYKJAVÍK · 2026

THE COMPANY

We build habits **worth having.**

Hörmann takes what game studios spent twenty years learning about attention, habit and delight — and points it at what a child actually needs.

Curiosity

over compliance. A child who wants to know beats a child who was told.

Effort

over talent. The game rewards the trying, never the being clever.

Failure

as information. Getting it wrong is a step in the method, not a verdict.

Games are the best teachers we have built. They are teaching the wrong things.

THE PREMISE

We already know how to make a child unable to stop.

42

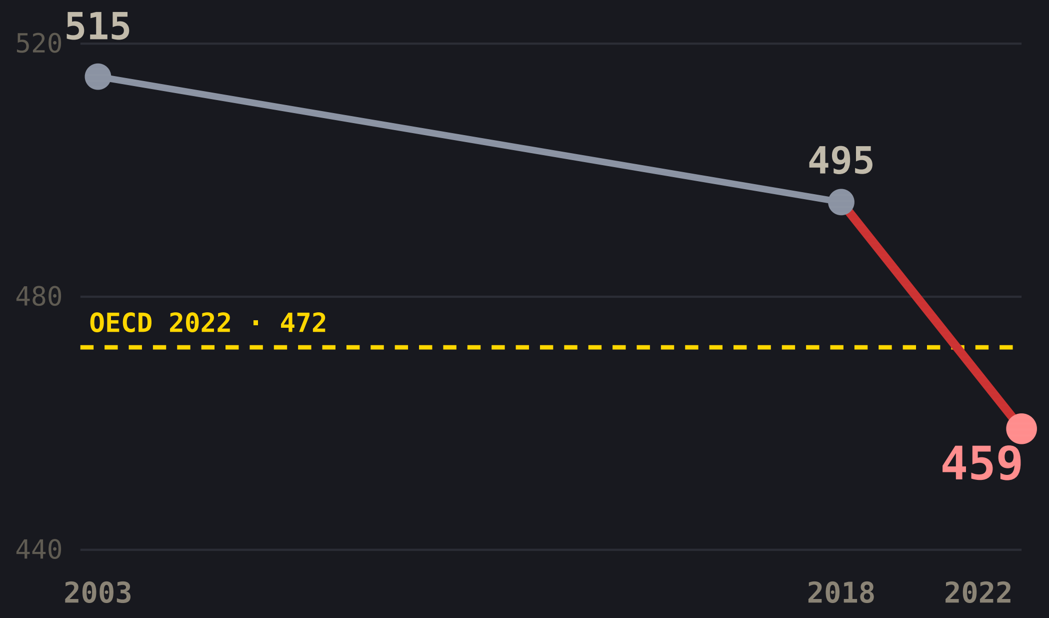
attorneys general are suing Meta for **deliberately designing addiction** into products used by children.[1]

It works. Nobody ever aimed it at anything good.

Iceland just posted its worst maths result ever.

34% of Icelandic 15-year-olds lack **basic maths proficiency.**[3]

Attention got better at holding them. School did not.



Iceland, PISA maths mean.[2] Axis 440–520.

OECD: the decline began before 2018, then accelerated.[2]

THE IDEA

Same craft. Better target.

WHAT THE ATTENTION INDUSTRY PERFECTED	WHAT A CHILD LEARNING MATHS NEEDS
▸ Feedback in under a second	▸ Feedback in under a second
▸ Always *just barely* winning	▸ Always *just barely* winning
▸ Reward that varies	▸ Reward that varies
▸ A reason to come back tomorrow	▸ A reason to come back tomorrow
▸ Never let them feel stupid	▸ Never let them feel stupid

The same list. Nobody aimed it at times tables.

Across 33 studies, game-based STEM teaching beats conventional instruction by ES 0.67.[4]

WHY ME

My son quits the second he smells failure.

ADHD, ODD, a high IQ. He will fight a hard boss for two hours. He will not do ten minutes of worksheets.

I make games. So I stopped shopping and started building.

Immediate, or nothing

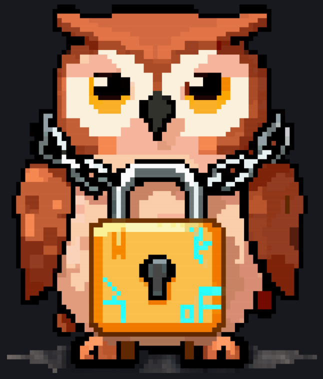
Children with ADHD need reinforcement that is **immediate, frequent and certain.** Delayed reward never bridges the gap.[5]

The clever bird frees the wise ones.

01 **Hörmann is a crow.** The cleverest bird alive — a tool-user, a face-reader, a solver of problems nobody set for him.

02 **The owls hold what is known.** Someone has put them in chains and walked away.

03 **A child who works something out sets one free.** That is the whole game, and the whole point.



The child is never the one being tested. The child is the one doing the rescuing.

HOW IT TEACHES

The maths is the gate, never the homework.

A world to move through

Run, jump, peck, collect. A real platformer first.

Every gate is a question

Answer it and the way opens. Refuse and you go nowhere.

Nothing is a quiz

There is no score, no marks, no report. Only an owl, freed.

Take the maths out and the game stops working.

THE SCIENCE

There is a right answer to how often a child should be right.

Right too often and nothing is learned. Right too rarely and they leave. Two independent lines of research land in the same window — and that window is the only thing our algorithm is trying to hit.

75%

the success rate the largest Elo-rated maths system targets. **50% is more informative but "discouragingly low".**[6]

85%

the optimal training accuracy derived for any gradient learner, human or machine. **Nature Communications.**[7]

70–85%

the band Hörmann holds a child inside, per domain, adjusting on **every single answer.**

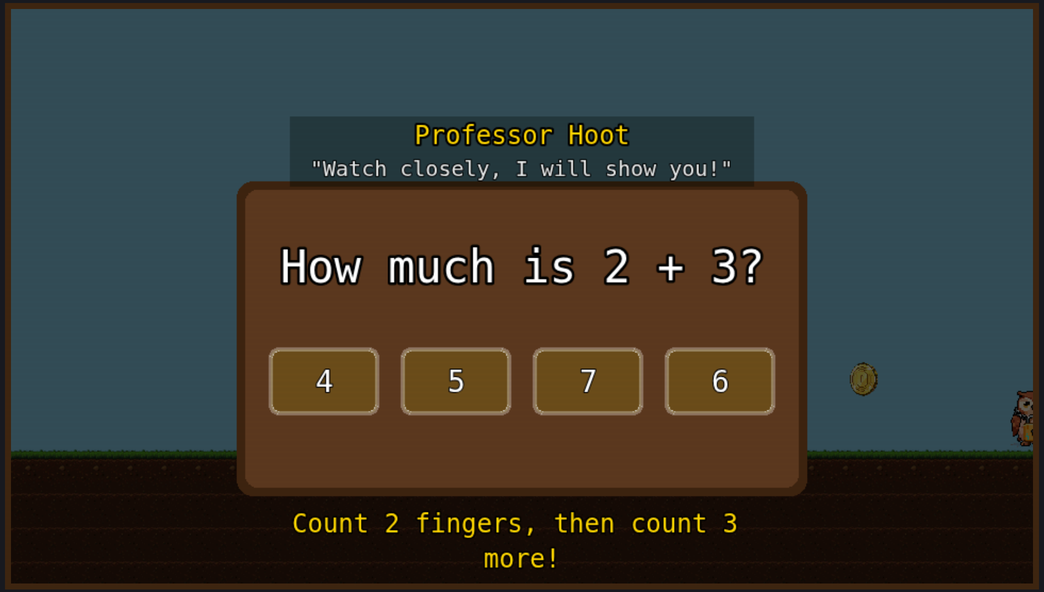
FAILURE

Celebrating failure is the whole trick.

01 **New maths is taught before it is asked.** The first attempt cannot hurt you.

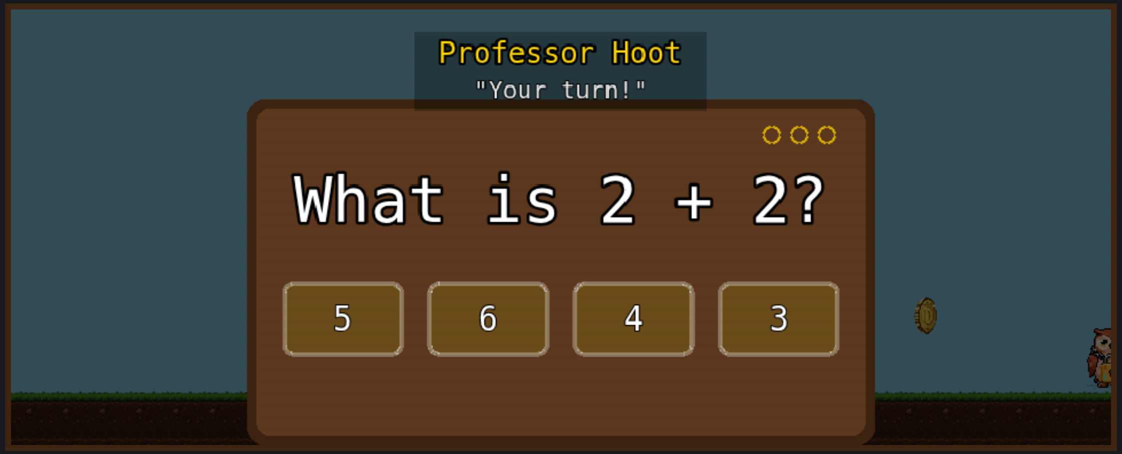
02 **A miss reveals the answer and explains it.** Getting it wrong is how you are taught.

03 **The comeback is the loudest moment in the game —** bigger than any first-time win.

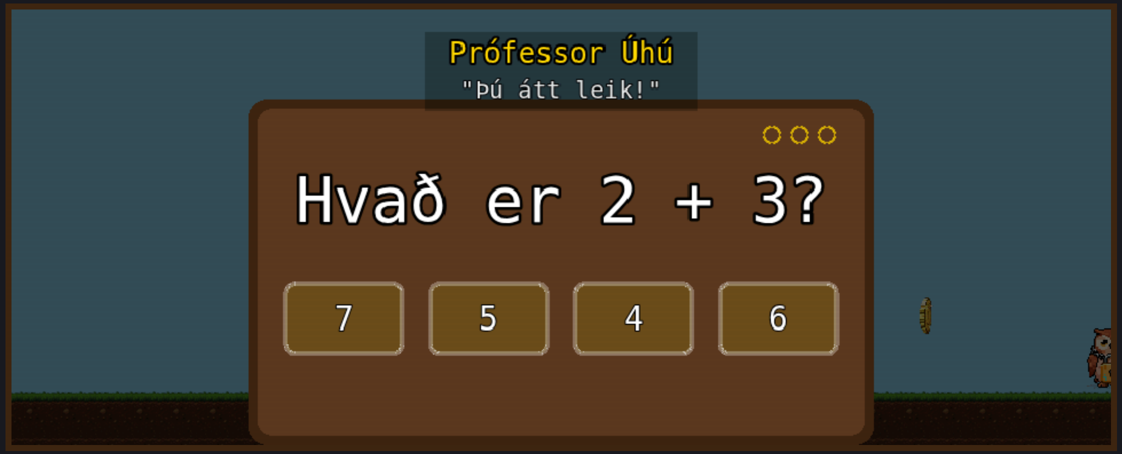


WHERE WE ARE

Not a concept. It runs.



LIVE ENCOUNTER · ENGLISH



SAME BUILD · ÍSLENSKA

Built

3,150 problems · 6 levels · Elo learner model · per-child profiles · two languages.

Played

One child, near-daily, for months. **n=1, and we say so.** No telemetry yet.

Next

One Icelandic class, playing and giving feedback. [pilot status]

Every screenshot in this deck came out of the running build.

HOW IT GROWS

The game does the selling.

Nobody markets to a child. The child does the marketing.

THE MODEL

Free while we prove it.
Then both sides pay.

PARENTS · B2C	SCHOOLS · B2B
▸ Home subscription, low price, high volume	▸ Per-pupil annual licence — the durable revenue
▸ Signal in weeks, not procurement quarters	▸ Every licence hands us a classroom of parents

TODAY
Free. Prototyping and feedback.

DELIVERY
Web. No COGS, no shipping.

PRICING
2,000 ISK per pupil / year · $10 per family / month

Each channel sells the other.

Price is set after the efficacy study, not before it.

The maths leaders are worksheets with sprites.

Prodigy

$125m raised, tens of millions of learners — an RPG wrapper around a question bank.[8]

SplashLearn

~1,900 activities, pre-K to grade 5. Breadth over depth, and worksheets in the box.[8]

DragonBox

Genuinely good puzzle design, ages 4–14 — but a puzzle set, not a world kids return to.[8]

None of them is a game you would play if the maths were removed.

Break-even is 6,500 families.

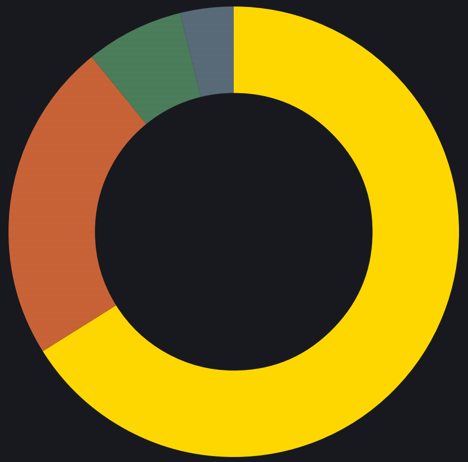
WHAT IT COSTS TO RUN	WHAT COVERS IT	WHAT IT LOOKS LIKE AFTER
€535k	**6,500 subs**	**3.1×**
A YEAR, ALL IN	AT $10 A MONTH	AT 20,000 SUBSCRIPTIONS
Two full-time people	Or **half of Iceland's schools + 2,400 families**	**$1.9m a year** against a $623k cost base
Contract art, audio and academic time	47,162 pupils sit in **174 schools**[9]	Every extra child is **~80% margin**

20,000 subscribers is not break-even. It is $1.3m a year.

Pupil and school counts: Statistics Iceland, 2024.[9] Prices are dials. ISK/EUR 141, ISK/USD 121.

THE ASK

€600k for twelve months.



- **66%** Staffing — two full-time seats
- **23%** Contracting — art, audio, academic
- **7%** Classroom pilot and efficacy study
- **4%** Platform, hosting, tools

On this we deliver a finished first world, a classroom, a study — and first revenue.

SOURCES

Every number, traceable.

1 42 state and territory attorneys general v. Meta, filed Oct 2023 — a 33-state joint federal suit plus nine individual filings. njoag.gov

2 OECD, PISA 2022 Results — Country Note: Iceland. Maths mean 459 (2022), 495 (2018), 515 (2003); OECD average 472; decline predates 2018. oecd.org

3 Miðstöð menntunar og skólaþjónustu, Helstu niðurstöður PISA 2022 á Íslandi. 66% at or above basic maths proficiency. mms.is

4 Wang et al., Effects of digital game-based STEM education on students' learning achievement: a meta-analysis, Int. J. STEM Educ., 2022. 33 studies, N=3,894, ES 0.667. doi 10.1186/s40594-022-00344-0

5 Luman et al., Reinforcement Contingency Learning in Children with ADHD, Res. Child Adolesc. Psychopathol., 2019. doi 10.1007/s10802-019-00572-z

6 Klinkenberg, Straatemeier & van der Maas, Computer adaptive practice of Maths ability…, Computers & Education 57(2), 2011. Math Garden targets a .75 success probability; .50 is more informative but "discouragingly low". 3,648 children, 3.5m problems in ten months, ~33% outside school hours. doi 10.1016/j.compedu.2011.02.003

7 Wilson, Shenhav, Straccia & Cohen, The Eighty Five Percent Rule for optimal learning, Nature Communications 10:4646, 2019. A machine-learning result, cited here as the upper bound of the same window. doi 10.1038/s41467-019-12552-4

8 Competitor figures: Prodigy funding via Tracxn company profile; SplashLearn activity count and reach via its own published materials; DragonBox age range via publisher. Public sources, not independently audited.

9 Statistics Iceland, Pupils in compulsory schools 2024: 47,162 pupils, 174 schools. statice.is

· Product figures — 3,150 problems, 6 levels, Elo learner model, the 70–85% band, EN/IS — come from the Hörmann repository's dated snapshot and architecture docs, verified against the running build. Play counts are stated as n=1 because that is what they are.